

- Description

This module manages a smart washing machine system, ensuring that:

1. The machine continuously monitors the weight of the laundry load (in kg).
2. If the load exceeds the maximum capacity, a warning is triggered and the washing cycle is not started.
3. If the load is within capacity, the system calculates the optimal cycle duration (in minutes) based on the load weight and a base cycle time.
4. The water consumption is also computed using a numerical formula based on the load weight.

This system helps optimize washing performance, prevent machine overloading, and reduce water and energy consumption.

- Specification

module SmartWashingMachineControl

var currentLoad, maxCapacity, baseCycleTime, waterPerKg, calculatedCycleTime, calculatedWaterUsage: real

process setLoad(load: real) returns (calculatedCycleTime: real, calculatedWaterUsage: real)

functional scenario form:

currentLoad <= maxCapacity && calculatedCycleTime = baseCycleTime + (currentLoad / maxCapacity) * baseCycleTime, calculatedWaterUsage = currentLoad * waterPerKg, output "Load accepted: " + currentLoad + " kg, Cycle Time: " + calculatedCycleTime + " minutes, Water Usage: " + calculatedWaterUsage + " liters", Optimal cycle and water usage calculated.

||

currentLoad > maxCapacity && calculatedCycleTime = 0, calculatedWaterUsage = 0, output "WARNING: Load exceeds maximum capacity.", Overload warning triggered, outputs zero values.

end process

end module

- Java Code

```
package code;
class c72_SmartWashingMachine {
    private double maxCapacity;
    private double baseCycleTime;
    private double waterPerKg;

    private double currentLoad;
    private double calculatedCycleTime;
    private double calculatedWaterUsage;
```

```

    public c72_SmartWashingMachine(double maxCapacity, double baseCycle
Time, double waterPerKg) {
        this.maxCapacity = maxCapacity;
        this.baseCycleTime = baseCycleTime;
        this.waterPerKg = waterPerKg;
        this.currentLoad = 0.0;
        this.calculatedCycleTime = 0.0;
        this.calculatedWaterUsage = 0.0;
    }

    public void setLoad(double load) {
        currentLoad = load;
        if (currentLoad > maxCapacity) {
            System.out.println("WARNING: Load exceeds maximum capacity!
Please reduce the laundry load.");
            calculatedCycleTime = 0.0;
            calculatedWaterUsage = 0.0;
        } else {
            calculatedCycleTime = baseCycleTime + (currentLoad / maxCap
acity) * baseCycleTime;
            calculatedWaterUsage = currentLoad * waterPerKg;
            System.out.printf("Load accepted: %.2f kg\n", currentLoad);
            System.out.printf("Calculated Cycle Time: %.2f minutes\n",
calculatedCycleTime);
            System.out.printf("Estimated Water Usage: %.2f liters\n", c
alculatedWaterUsage);
        }
    }

    public double getCycleTime() {
        return calculatedCycleTime;
    }

    public double getWaterUsage() {
        return calculatedWaterUsage;
    }

    public static void main(String[] args) {
        c72_SmartWashingMachine machine = new c72_SmartWashingMachine(8.0,
30.0, 10.0);

        // testcase-VT:
        //     machine.setLoad(5.0);
        //     machine.setLoad(9.0);

```

```
//      machine.setLoad(7.0);

//      // testcase-FT:
//      machine.setLoad(-1.7836486571579143E308);
//      machine.setLoad(0.0);
//      machine.setLoad(1.6757390077074176E303);

//      // testcase-Z3:
//      machine.setLoad(5.0);
//      machine.setLoad(9.0);
//      machine.setLoad(7.0);

//      // testcase-UVT:
//      machine.setLoad(5.0);
//      machine.setLoad(9.0);
//      machine.setLoad(7.0);
    }
}
```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c72_SmartWashingMachine.java	100.0 %	2	0	2
✓ c72_SmartWashingMachine	100.0 %	2	0	2
main(String[])		0	0	0
c72_SmartWashingMachine(dou		0	0	0
setLoad(double)	100.0 %	2	0	2

3. Vibration testing instruction coverage: 95%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	114	20,823	20,937
✓ src	0.5 %	114	20,823	20,937
✓ code	0.5 %	114	20,823	20,937
✓ c72_SmartWashingMachine.java	95.0 %	114	6	120
✓ c72_SmartWashingMachine	95.0 %	114	6	120
main(String[])	100.0 %	17	0	17
c72_SmartWashingMachine(dou	100.0 %	21	0	21
setLoad(double)	100.0 %	76	0	76

4. Fuzz testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c72_SmartWashingMachine.java	100.0 %	2	0	2
✓ c72_SmartWashingMachine	100.0 %	2	0	2
main(String[])		0	0	0
c72_SmartWashingMachine(dou		0	0	0
setLoad(double)	100.0 %	2	0	2

5. Fuzz testing instruction coverage: 95 %

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	114	20,823	20,937
✓ src	0.5 %	114	20,823	20,937
✓ code	0.5 %	114	20,823	20,937
✓ c72_SmartWashingMachine.java	95.0 %	114	6	120
✓ c72_SmartWashingMachine	95.0 %	114	6	120
main(String[])	100.0 %	17	0	17
c72_SmartWashingMachine(dou	100.0 %	21	0	21
setLoad(double)	100.0 %	76	0	76

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c72_SmartWashingMachine.java	100.0 %	2	0	2
✓ c72_SmartWashingMachine	100.0 %	2	0	2
main(String[])		0	0	0
c72_SmartWashingMachine(dou		0	0	0
setLoad(double)	100.0 %	2	0	2

7. Z3 instruction coverage: 95%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	114	20,823	20,937
✓ src	0.5 %	114	20,823	20,937
✓ code	0.5 %	114	20,823	20,937
✓ c72_SmartWashingMachine.java	95.0 %	114	6	120
✓ c72_SmartWashingMachine	95.0 %	114	6	120
main(String[])	100.0 %	17	0	17
c72_SmartWashingMachine(dou	100.0 %	21	0	21
setLoad(double)	100.0 %	76	0	76

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
✓ VibrationTestingTool	0.2 %	2	840	842
✓ src	0.2 %	2	840	842
✓ code	0.2 %	2	840	842
✓ c72_SmartWashingMachine.java	100.0 %	2	0	2
✓ c72_SmartWashingMachine	100.0 %	2	0	2
main(String[])		0	0	0
c72_SmartWashingMachine(dou		0	0	0
setLoad(double)	100.0 %	2	0	2

9. Unimproved V-method instruction coverage: 95%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
✓ VibrationTestingTool	0.5 %	114	20,823	20,937
✓ src	0.5 %	114	20,823	20,937
✓ code	0.5 %	114	20,823	20,937
✓ c72_SmartWashingMachine.java	95.0 %	114	6	120
✓ c72_SmartWashingMachine	95.0 %	114	6	120
main(String[])	100.0 %	17	0	17
c72_SmartWashingMachine(dou	100.0 %	21	0	21
setLoad(double)	100.0 %	76	0	76